

Unstable Bradycardia



Section I: Scenario Demographics

Scenario Title:	Unstable Bradycardia		
Date of Development:	18/07/2015		
Target Learning Group:	☒ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Martin Kuuskne
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To manage a patient with unstable bradycardia leading to cardiac arrest
CRM Objectives:	1) Predicts the likelihood of deterioration in a patient with bradycardia and describes a plan of management to team members. 2) Recognizes the need to effectively utilize resources by optimizing pharmacologic therapy and instituting transcutaneous pacing in addition to early consultation with cardiology for placement of an intravenous pacemaker.
Medical Objectives:	1) Employ a intravenous chronotropic agent for unstable bradycardia 2) Promptly recognize pulseless electrical activity in the setting of an unstable patient.

Case Summary: Brief Summary of Case Progression and Major Events
A 78-year-old male presents to the emergency department with an unstable bradycardia. The patient deteriorates from a second degree, Mobitz Type II-AV block into a third degree AV block requiring ACLS protocol medications, transcutaneous pacing, and ultimately transvenous pacing until definitive management with a permanent pacemaker can be arranged.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby

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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
A 78-year-old male from a long-term care facility is being transferred to the emergency department for decreased mental status.			
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
N/A	N/A		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☒ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
Clothing/attire to fit patient age, elderly mask on the mannequin if available.			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	10 min
		Debriefing:	10 min

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Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: Gabriel DeFabio		Age: 78	Weight: 100kg
Gender: ☒ M ☐ F		Code Status: Full-Code	
Chief Complaint: Altered Mental Status			
History of Presenting Illness: The patient was found at his residence with an altered mental status. The caretakers noticed "abnormal movements" twitching of his head and the patient was not responding appropriately. There was no history of chest-pain or shortness of breath prior to the episode.			
Past Medical History:	CAD with remote MI	Medications:	ASA 80mg PO Daily
	HTN		Bisoprolol 5mg PO Daily
	Hypothyroidism		Levothyroxine 0.88mg PO Daily
	Gout		Colchicine 0.6mg PO daily
Allergies: Penicillin			
Social History: non contributory			
Family History: non contributory			
Review of Systems:	CNS:	Patient has a decreased mental status	
	CVS:	No chest pain	
	RESP:	No SOB	
	GI:	No GI changes	
	GU:	No GU changes	
	MSK:	No Complaints	INT:
B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor On, no data displayed	
☐ Monitor on Standard Display			
HR: 40/min	BP: 97/63	RR: 12/min	O ₂ SAT: 91%
Rhythm: 2°AVB Mobitz II	T: 35.9°C	Glucose: 7.0 mmol/L	GCS: 9 (E2V3M4)
General Status: Decreased mental status			
CNS:	Pearl, withdrawing all limbs to pain.		
HEENT:	Normal		
CVS:	Bradycardia		
RESP:	GAEB		
ABDO:	Soft, non-tender		
GU:	Normal		
MSK:	Normal	SKIN:	Normal, cap refill >3 seconds.

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Section VI: Scenario Progressi

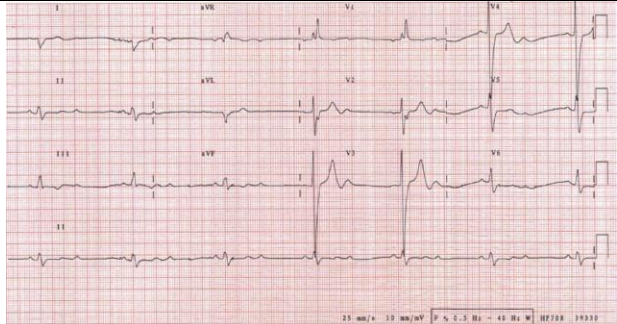
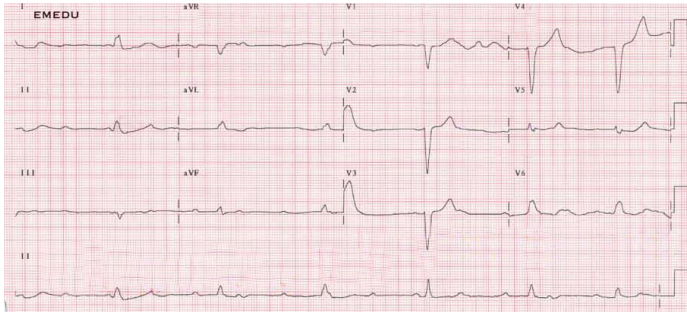
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: 2°AVB Mobitz II HR: 40/min BP: 97/63 RR: 12/min O ₂ SAT: 91% T: 36.1°C	Decreased mental status, GCS: 9	<u>Learner Actions</u> - ☐ Attaches monitors - ☐ Acquires IV access - ☐ Supplemental O ₂ - ☐ Hx/Physical Exam - ☐ EKG/CXR - ☐ Lab investigations - ☐ Recognizes unstable brady - ☐ Atropine IV 0.5mg - ☐ IV NS Bolus	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Supplemental O ₂ → O ₂ SAT to 95% <u>Triggers</u> <i>For progression to next state</i> - 3 minutes → 2. 3° AVB
2. 3°- AVB → Rhythm: 3°AVB → HR: 26 → BP: 61/42	Decreased mental status, GCS: 9	<u>Learner Actions</u> - ☐ Atropine IV 0.5mg - ☐ Dopamine/Epinephrine infusion - ☐ Isoproterenol Infusion - ☐ EKG - ☐ Transcutaneous Pacing - ☐ IV Analgesic for pacing	<u>Modifiers</u> - IV chronotropic agent → HR to 40, BP:70/45 - Transcutaneous pacing → No mechanical capture until 70mA <u>Triggers</u> - 3 minutes without pacing → 3. PEA - If begins pacing → 4. Paced Rhythm
3. PEA → BP: 0/0 → RR: 0 → O ₂ SAT: 0%	Non-responsive	<u>Learner Actions</u> - ☐ Chest compressions - ☐ 1mg Epinephrine IV - ☐ Crash intubation	<u>Modifiers</u> Intubation → Display FiO ₂ = 1.0 T _v = 500 PEEP = 5 T _v = 500 mL <u>Triggers</u> - 2 Rounds of CPR → 3. 3°-AVB
4. Paced Rhythm Rhythm → 3°AVB HR → Paced Rhythm at decided rate BP: 100/65 → O ₂ SAT: 90% → RR: 12	Decreased mental status, GCS: 9 if not intubated	<u>Learner Actions</u> - ☐ Intubation (if not done during code) - ☐ Transcutaneous Pacing - ☐ Transvenous Pacing - ☐ Cardiology consult - ☐ Repeat EKG	END SCENARIO PRN.

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na:	K:	Cl:	HCO ₃ :	BUN:	Cr:	Glu:
Ca:		Mg:		PO ₄ :		Albumin:
VBG	pH:		PCO ₂ :	PO ₂ :	HCO ₃ :	Lactate:
WBC:		Hg:		Hct:		Plt:
No Laboratory Results available for this case						

Images (ECGs, CXRs, etc.)	
 2nd Degree AVB Mobitz 2	
 3rd Degree AVB	

Ultrasound Video Files (if applicable)	
U/S Showing non-capture	U/S Showing Capture



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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Medical Objectives:	1) Employ a intravenous chronotropic agent for unstable bradycardia 2) Promptly recognize pulseless electrical activity in the setting of an unstable patient.
Sample Questions for Debriefing	
1) How to you prepare the team for potential deterioration of the patient? 2) Why does Atropine tend not to improve in high-degree AV Blocks? 3) What are ways to check for electromechanical capture during pacing? 4) How to you start and maintain pacing for symptomatic bradycardia? 5) What is the ultimate disposition for patients with 3 rd degree AVB?	
Key Moments	
1) Deterioration from 2 nd to 3 rd degree AVB	
2) PEA arrest	
3) Electromechanical Capture	